AISC Practical Exam – Theory for Experiments 2–9

Experiment 2: Breadth-First Search (BFS) for the Water Jug Problem

Theory:

Breadth-First Search (BFS) is an uninformed search algorithm used to find a path from an initial state to a goal state. It explores all states at the current level before moving to the next level. BFS uses a queue (FIFO – First In, First Out) to store states that need to be explored.

In the Water Jug Problem, two jugs of different capacities are used to measure a required quantity of water. Each state represents the amount of water present in both jugs. The possible actions are filling a jug, emptying a jug, or pouring water from one jug into the other.

BFS explores all possible states systematically until the goal state is reached. If a solution exists, BFS finds the shortest path in terms of the number of actions when each action has equal cost. It is simple and complete but may require more memory because it stores many states.

Experiment 3: A* Search for the Water Jug Problem

Theory:

A* Search is an informed search algorithm that finds a path from an initial state to a goal state using the cost of reaching a state and an estimate of the remaining cost. It uses the evaluation function:

$$f(n) = g(n) + h(n)$$

Here, $g(n)$ is the actual cost from the starting state to the current state, and $h(n)$ is the estimated cost from the current state to the goal.

In the Water Jug Problem, each state represents the amount of water in the two jugs. The algorithm generates possible states by filling, emptying, or pouring water between the jugs. It selects the state with the lowest $f(n)$ value for further exploration.

A* Search can find an optimal solution when the heuristic is appropriate, particularly when it does not overestimate the remaining cost. Compared with BFS, it uses additional information to guide the search toward the goal.

Experiment 4: Minimax Algorithm with Alpha-Beta Pruning

Theory:

The Minimax algorithm is a search technique used in Artificial Intelligence for two-player games such as chess and tic-tac-toe. It assumes that both players play optimally. One player, called MAX, tries to obtain the highest possible score, while the other player, called MIN, tries to obtain the lowest possible score.

The algorithm represents possible moves using a game tree. It evaluates the tree from the leaf nodes toward the root and selects the best move based on the scores of the possible outcomes.

Alpha-Beta Pruning improves the efficiency of Minimax by avoiding the evaluation of branches that cannot affect the final decision. Alpha represents the best value found so far for MAX, and Beta represents the best value found so far for MIN. When alpha becomes greater than or equal to beta, the remaining branches can be pruned.

Alpha-Beta Pruning produces the same final decision as Minimax but can reduce the number of nodes evaluated, making the search faster.

Experiment 5: Genetic Algorithm for Function Optimization

Theory:

A Genetic Algorithm (GA) is an optimization technique inspired by the process of natural selection. It is used to find good or optimal solutions to complex problems by improving a population of candidate solutions over multiple generations.

Each candidate solution is called a chromosome, and its individual components are called genes. A fitness function measures how good each candidate solution is.

The main steps of a Genetic Algorithm are:

1. Initialization: Generate an initial population of candidate solutions.
2. Fitness evaluation: Calculate the fitness of each solution.
3. Selection: Select suitable solutions to become parents.
4. Crossover: Combine parts of two parents to produce new solutions.

5. Mutation: Make small random changes to maintain diversity.
6. Replacement: Form a new population and repeat the process.

The algorithm continues until a stopping condition is met, such as reaching a maximum number of generations or finding a satisfactory solution. Genetic Algorithms are useful for optimization problems where checking every possible solution is difficult.

Experiment 6: Single-Layer Perceptron (SLP) Using the Perceptron Learning Rule

Theory:

A Single-Layer Perceptron (SLP) is a simple Artificial Neural Network used for binary classification. It consists of input values, weights, a bias, a summation function, and an activation function.

Each input is multiplied by its corresponding weight. The weighted inputs and bias are added to calculate the net input:

$$net = w_1 x_1 + w_2 x_2 + b$$

A step activation function converts the net input into an output of either 0 or 1. The perceptron compares this output with the target value and calculates the error:

$$Error = Target - Output$$

The weights and bias are updated using the perceptron learning rule:

$$w_i(new) = w_i(old) + \eta(Error)x_i$$

$$b(new) = b(old) + \eta(Error)$$

Here, η is the learning rate. The process repeats until the perceptron correctly classifies the training samples or reaches the maximum number of training iterations.

In this experiment, the perceptron is trained to implement the AND logic gate. It learns to output 1 only when both inputs are 1. A single-layer perceptron can learn linearly separable patterns such as AND and OR, but it cannot independently learn the XOR function.

Experiment 7: Supervised Learning Using Multi-Layer Perceptron (MLP)

Theory:

A Multi-Layer Perceptron (MLP) is an Artificial Neural Network that consists of an input layer, one or more hidden layers, and an output layer. It is used for supervised learning, where the network learns from input data paired with known target outputs.

Each neuron receives inputs, multiplies them by weights, adds a bias, and applies an activation function to produce an output. The information moves forward from the input layer through the hidden layers to the output layer. This process is called forward propagation.

The network compares its predicted output with the target output to calculate the error. During training, backpropagation calculates how the weights contributed to the error, and an optimization method updates the weights and biases to reduce it.

An MLP can learn complex, non-linear relationships. Unlike a single-layer perceptron, an MLP with a suitable hidden layer and activation functions can learn the XOR logic gate. Training continues until the error is sufficiently reduced or a stopping condition is reached.

Experiment 8: Unsupervised Learning Using K-Means Clustering

Theory:

K-Means is an unsupervised machine-learning algorithm used to group similar data points into clusters. It does not require predefined class labels. Instead, it identifies groups based on the similarity or distance between data points.

The value K represents the number of clusters to be formed. The algorithm begins by selecting K initial cluster centroids. Each data point is assigned to the nearest centroid, usually using Euclidean distance.

After assigning the data points, the algorithm recalculates each centroid by taking the mean of all points belonging to that cluster. The assignment and centroid-update steps are repeated until the centroids stop changing significantly or a maximum number of iterations is reached.

K-Means aims to minimize the sum of squared distances between each data point and its assigned cluster centroid. It is commonly used for customer segmentation, grouping similar records, and exploratory data analysis. Its results can depend on the initial centroid positions and the selected value of K.

Experiment 9: Fuzzy Membership Functions

Theory:

Fuzzy Logic is a method of reasoning that represents degrees of truth between 0 and 1 instead of using only the exact values 0 and 1 of traditional Boolean logic. It is useful when information is uncertain or concepts are not clearly defined.

A fuzzy membership function assigns a membership value to an input, indicating how strongly that input belongs to a fuzzy set. A membership value of 0 means no membership, 1 means full membership, and a value between 0 and 1 represents partial membership.

A triangular membership function is defined by three parameters: a, b, and c. These represent the left endpoint, peak, and right endpoint of the triangle. Its membership value is given by:

\[\mu(x)= \begin{cases} 0, & x\leq a\\ \dfrac{x-a}{b-a}, & a<x\leq b\\ \dfrac{c-x}{c-b}, & b<x<c\\ 0, & x\geq c \end{cases} \]

The membership value increases from 0 to 1 between a and b, reaches 1 at b, and decreases from 1 to 0 between b and c.

Fuzzy membership functions are used in fuzzy control systems, decision-making, and applications involving linguistic concepts such as low, medium, and high temperature. They help represent gradual transitions instead of strict boundaries.